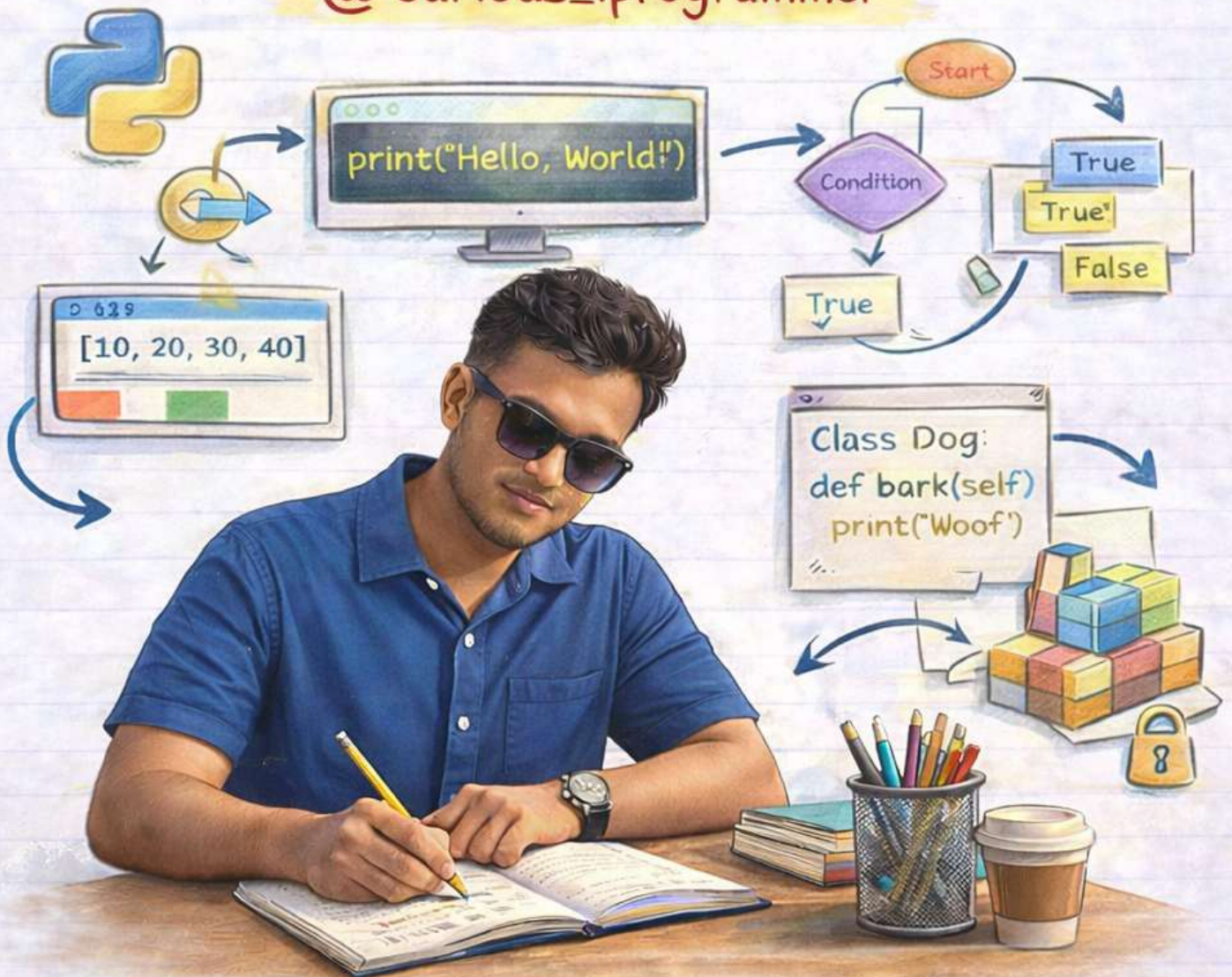


Complete PYTHON Notes

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Chapter 1: Introduction to Python

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What is Python?

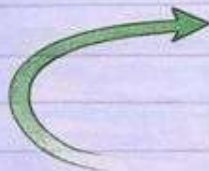
Python is a popular, high-level, interpreted programming language known for its simplicity and readability. It is used for web development, data science, AI, automation, and more.



History of Python



Guido van Rossum



- **1989**: Python created by Guido van Rossum at CWI, Netherlands.
- **1991**: Python 0.9.0 released - included features like classes, functions.
- **2000**: Python 2.0 released - added list comprehensions, garbage collection.
- **2008**: Python 3.0 released - improved language constructs, better Unicode support.

Features of Python



Easy to Learn: Simple and easy to-understand syntax



Readable: Code is clear and readable like English



Interpreted: Executes code line by line, no need for compilation.



Cross-Platform: Runs on various platforms (Windows, Mac, Linux)



Cross-Platform

Runs on various platforms (Windows, Mac, Linux)



Large Standard Library

Rich set of modules & functions for various tasks



Dynamically Typed

No need to declare variable types explicitly



Open Source

Free and open source, with a large community

Advantages & Disadvantages

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✓ Advantages

- ✓ **Simple and Easy to Learn:**
Reads like English, reducing learning curve.
- ✓ **Extensive Libraries:** Rich set of libraries for various tasks.
- ✓ **Versatile:** Used in web development, data science, AI,
- ✓ **Great Community Support:**
Large, active community with plenty of resources.
- ✓ **Cross-Platform:** Runs on Windows, Mac, Linux, etc.
- ✓ **High-Level Language:**
Easy to write and understand.

✗ Disadvantages

- ✗ **Slower Execution:**
Interpreted nature makes it slower than compiled languages.
- ✗ **Memory Consumption:**
High memory usage due to dynamic nature.
- ✗ **Not Ideal for Mobile:**
Less suitable for mobile app development.
- ✗ **GIL (Global Language:**
Limits performance in multi-threaded environments

Python vs Other Languages

Python

- ✓ **Simpler, Easy-to-Learn Syntax**
- ✓ **Interpreted, Dynamic Typing**
- ✓ **Large Standard Library**
- ✓ **Great Libraries for AI, Data Science, Web Dev**

Other Languages

- ✓ **Faster Execution Speed**
- ✓ **Statically Typed Languages**
- ✓ **Better for Mobile Development**
- ✓ **More Control Over Memory Usage**